

Separating compression, likelihood approximation and dispersive response errors in simulated pulsar timing arrays

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Abstract

Inference from stochastic backgrounds involves distinct choices of frequency compression and likelihood approximation. We study their separation in a periodic Fourier experiment for a dispersive tensor background in pulsar timing arrays. The comparison retains complex coefficients, constructs quadratic estimators, compresses their means and covariances consistently, and changes the response while holding the compressed observation fixed. An exactly Gaussian control isolates information loss from the non-Gaussian distribution of the physical quadratic estimators. A final campaign uses two coupled configurations, with 12 pulsars and four frequencies or 16 pulsars and eight frequencies, and 596 realizations per configuration, yielding 10728 conditional posterior analyses. Sky directions are taken from public MeerKAT metadata; all distances, signals and noise are synthetic. Interval-sensitive calibration tests retain numerically unresolved targets. Separate Holm families contain 42 correctly specified and 84 approximate-model tests. They yield respectively zero and one persistent rejection, with three and one indeterminate decisions. For 500 prior draws, the paired mean information loss from Gaussian compression is 0.06211 ± 0.00477 and 0.10631 ± 0.00695 nat in the two configurations, where uncertainties are Monte Carlo standard errors. The results separate compression loss from likelihood misspecification. A constant likelihood provides a useful baseline: the uniform prior alone yields a 95th-percentile upper limit of $u = 0.95$ with zero information gain. Mass limits therefore require an assessment of posterior learning as well as calibration.

1 Introduction

A dispersive gravitational-wave response depends on both frequency and pulsar geometry. Replacing it by a response at one reference frequency is therefore a model approximation whose inferential consequences need to be tested. However, comparisons may simultaneously change the retained data, the likelihood family and the response itself. A difference between resulting mass posteriors cannot then be attributed uniquely to frequency compression.

Full-covariance, multifrequency graviton-mass forecasts are already available, including PTA-astrometry estimators [1]. The breakdown of Gaussian approximations to stochastic-background likelihoods and its connection to compression have also been studied [2]. Building on these results, we compare dispersive PTA responses using matched observations and separate controls for compression and likelihood approximation. Propagating numerical uncertainty through the calibration tests reveals which differences can be attributed to the statistical model.

The numerical experiment conditions on known noise parameters and uses periodic Fourier coefficients. This setting supplies an explicit data-generating distribution and isolates the effects of the approximations. We study two coupled array configurations and relate their calibration and information gain to the complementary experiments in the companion dissertation.

2 Experiment and comparison design

2.1 Periodic data and dispersion

For positive Fourier frequencies $f_k = k/T$, let q_k be a proper complex Gaussian vector,

$$q_k \sim \mathcal{CN}(0, C_k(u)), \quad C_k = P_{\text{GW}}(f_k)\Gamma_k(u) + N_k, \quad u = f_g T \in [0, 1]. \quad (1)$$

Frequencies are independent by definition of the periodic experiment. The dispersion variable is $\beta_k = \sqrt{1 - (u/k)^2}$ and the mass conversion is $m_g c^2 = hu/T$. The prior is uniform in u . This finite support is a modeling choice, not an empirical bound. The response includes Earth and pulsar terms, evaluated with harmonic expansions and independent sky-integration checks, following the dispersive PTA setting of Refs. [3, 4]. With transverse tensor polarizations of norm two, the convention is

$$R_a^A = \frac{p_a^i p_a^j e_{ij}^A}{2(1 + \beta \hat{\Omega} \cdot p_a)} [1 - e^{-iy_a(1 + \beta \hat{\Omega} \cdot p_a)}], \quad (2)$$

$$\Gamma_{ab} = \frac{3}{8\pi} \int d\Omega \sum_{A=+, \times} R_a^A R_b^{A*}. \quad (3)$$

Removable denominator singularities are evaluated by a sinc representation. The pulsar-term dependence is retained, rather than taking an infinite-distance limit throughout.

For a real quadratic statistic $X_i = q^\dagger H_i q$ with Hermitian H_i , the proper-complex Gaussian generator gives

$$\mu_i = \text{tr}(H_i C), \quad \Sigma_{ij} = \text{Re tr}(H_i C H_j C). \quad (4)$$

In each frequency there are ten estimators: real and imaginary cross powers averaged uniformly in four equal-count bins of ordered pair cosines, plus two auto-power bins sorted by nominal timing precision. Bin definitions use only fixed geometry and nominal errors. Thus A contains 40 or 80 real coordinates and B contains ten. Frequency compression is $B_i = K^{-1} \sum_k X_{ki}$ on the fixed, scaled coordinates described below; it is not fitted to the realization.

These moments do not make the joint distribution of X Gaussian. We define A0 as the complex-coefficient likelihood, and A as a normal likelihood for the quadratic estimators using these moments. B applies a fixed linear compression W to A, with $\mu_B = W\mu_A$ and $\Sigma_B = W\Sigma_A W^\top$.

The response controls use the same observed B. In C_β , the relative propagation-speed argument of the response is set to $\beta_1(u) = \sqrt{1 - u^2}$ in every channel while the physical channel-dependent phase scale $y_k = 2\pi f_k L/c$ is retained. In C_{full} , the entire response is evaluated at its first-channel value $\Gamma(\beta_1(u), y_1)$ for the same mass. Physical spectral and noise factors remain unchanged. These controls are specified approximations, not interchangeable definitions of every possible effective-frequency scheme. Means and covariances are recomputed for each response. Both controls retain mass dependence. The normalized densities used for A0 and for a real normal vector x are

$$\log p(q | u) = - \sum_k [n_p \log \pi + \log \det C_k + q_k^\dagger C_k^{-1} q_k], \quad (5)$$

$$\log p(x | u) = -\frac{1}{2} [d \log(2\pi) + \log \det \Sigma + (x - \mu)^\top \Sigma^{-1} (x - \mu)]. \quad (6)$$

For A the latter expression sums independent frequency blocks; for B and C it uses the compressed covariance. We do not compare absolute evidences across different observed data spaces.

2.2 Exactly Gaussian control

To separate compression from the quadratic distribution, write $C = LL^\dagger$ and $A_i = L^\dagger H_i L$. Let Z be a random Hermitian matrix with independent standard normal diagonal entries and

off-diagonal entries $(x + iy)/\sqrt{2}$, where x, y are independent standard normals. Define

$$Y_i = \text{tr}(H_i C) + \text{tr}(A_i Z). \quad (7)$$

Then Y is jointly real Gaussian with the same μ and Σ as X . The construction also preserves the cross-covariance between the two pulsar configurations by embedding both sets of estimator matrices in the same 16-pulsar space. The physical CN and Gaussian G experiments use independent latent banks conditional on the same true mass. Within each bank, the prescribed comparisons share observations.

2.3 Population design

We compare P12K4 and P16K8 at $T = 4.5$ years. Sixteen sky directions are selected deterministically from public MeerKAT timing-model metadata, using a greedy maximum-minimum angular separation rule with fixed tie-breaking; the first twelve define P12. This is not a representative random sample of the collaboration. Synthetic distances range from 100 to 500 light years, nominal timing precisions from 100 to 1000 ns, and red-noise multipliers from 0.5 to 2. The corresponding values for shared pulsars remain identical. The largest fL/c is approximately 889 cycles.

The central known nuisance parameters are

$$(\log_{10} A_{\text{GW}}, \gamma_{\text{GW}}, \log_{10} A_r, \log_{10} \text{EFAC}) = (-15, 13/3, -15.5, 0), \quad (8)$$

with individual red-noise index four. Nominal cadence determines the white-noise PSD, not an irregular sampling window. The one-sided residual PSD convention is

$$P(f; A, \gamma) = \frac{A^2}{12\pi^2 f_{\text{yr}}^3} \left(\frac{f}{f_{\text{yr}}} \right)^{-\gamma}. \quad (9)$$

Noise is diagonal with entries $r_a^2 P(f; A_r, 4) + 2(\text{EFAC} \sigma_a)^2 \Delta t$, where $\Delta t = T/[T/(14 \text{ days})]$. Numerically, q_k is divided by $\sqrt{s_k}$ and C_k by s_k , with $s_k = P(f_k; 10^{-14.7}, 13/3) + 2 \text{median}(\sigma_{16})^2 \Delta t$. This deterministic scale is shared by both configurations and enters the definition of their frequency averages. The arbitrary reference amplitude in s_k is not an inferred noise value.

Five hundred masses are independent draws from the prior. Three additional cells contain 32 realizations each: $u = 0$ and $u = 0.995$ under central noise, and $u = 0.5$ with $\log_{10} A_r = -14.7$. Fixed cells are excluded from simulation-based calibration (SBC) and prior-averaged information. There are nine analyses per configuration: A0/CN, A/CN, B/CN, C_β /CN, C_{full} /CN, A/G, B/G, C_β /G and C_{full} /G. Thus 596 paired realizations give 10728 posterior targets. All inference here is conditional on known nuisance parameters.

3 Numerical validation and statistical decisions

The geometry benchmark compares harmonic resolutions, direct sky integration, analytic limits, rotations and subsets. A rotation discrepancy confined to the diagonal was traced to rounding of the self-angle in a Legendre recurrence. Enforcing the exact identity $P_\ell(1) = 1$ only on the diagonal removes it without covariance repair or changing off-diagonal entries. The corrected maximum discrepancies are 4.80×10^{-12} between resolutions and 6.14×10^{-14} against direct integration.

A 16-realization pilot compares 288 conditional posteriors. Eighteen prospectively selected direct functional references cover all model-configuration pairs. Quantile and event references pass after documented refinements; five event references require narrower boundary regions. The production grid has 971 mass nodes, with two harmonic resolutions. Positive Gauss-Legendre rules of orders 32 and 64 in $\alpha = \arcsin u$ are compared on fixed panels, with positive interpolated

densities and direct truth likelihoods. The numerical accuracy criteria require log-likelihood resolution changes, log-normalization, moment/KL changes and quantile differences at most 10^{-3} , CDF differences at most 2×10^{-3} , and a refined Wasserstein check. Mass-PIT bounds add 0.002 to the range of the two CDF evaluations. Event boundaries are enclosed by bands of half-width 10^{-5} in u around the roots of both finite interpolants. Their ambiguous probability mass is propagated; if the intrinsic event interval exceeds 0.002, the event remains unresolved. Accepted event bounds receive an additional 0.002 margin. These sensitivity intervals are supported by the direct references and finite-resolution comparisons in the sampled domain.

SBC validates the joint simulation–inference procedure, and its sensitivity depends on the test quantity [5, 6]. For each target, the mass PIT and the posterior probability of the likelihood-level event at the true parameter are propagated as intervals. Unresolved events receive $[0, 1]$, without dropping realizations. KS statistics and coverage counts are bounded over these intervals. We use four one-sided coverages, central 90% coverage, and KS tests for the mass and log-likelihood PITs. Separate Holm corrections are applied to 42 tests from correctly specified models (A0/CN, A/G and B/G in both configurations) and 84 approximate-model tests. Familywise error is controlled separately within these two groups.

A rejection is persistent when even its upper adjusted p bound is at most 0.05. A decision is indeterminate when rejection is permitted by the lower bound but not forced by the upper bound. Failure to reject is not evidence of equivalence. Generation is independently replayed from nominal seeds and stored RNG states. An independent aggregation implementation reconstructs coverage counts and Holm decisions from the retained functional outputs; it does not rerun physical integration. Both aggregations use SciPy for the KS survival function.

4 Results

All 10728 conditional posteriors were evaluated. The mass PITs satisfy the numerical accuracy criteria throughout the sample. Some likelihood-level events remain unresolved, reflecting their greater sensitivity to boundaries of the integration domain.

Table 1: Interval-sensitive SBC decisions, with separate 5% Holm corrections.

Family	Tests	Persistent	Indeterminate
Correctly specified	42	0	3
Approximate	84	1	1

The persistent rejection is the log-likelihood PIT KS test for A/CN in P12K4, whose adjusted upper p bound is 0.004956. The corresponding P16K8 test is indeterminate. The correctly specified indeterminate tests are A0/CN in both configurations and A/G in P12K4, all for log-likelihood PITs. Mass-PIT tests exclude rejection under the adopted adjusted interval bounds. The remaining indeterminacy is concentrated in the likelihood-level events.

For the correctly specified Gaussian control, the paired A/G minus B/G mean KL is 0.06211 ± 0.00477 nat in P12K4 and 0.10631 ± 0.00695 nat in P16K8. Because B is a fixed compression of A, the prior-average difference estimates information loss. The reported uncertainty reflects simulation variation, not systematic numerical error. Individual realizations need not exhibit this ordering. For a correctly specified model and deterministic $B = WA$, the identity

$$\mathbb{E}_A D_{\text{KL}}[p(u \mid A) \parallel p(u)] - \mathbb{E}_B D_{\text{KL}}[p(u \mid B) \parallel p(u)] = I(u; A \mid B) \geq 0 \quad (10)$$

follows from the chain rule of mutual information. Equality requires conditional sufficiency, $u \perp A \mid B$. This identity does not apply to approximate posterior families as though they were generated by the true joint distribution.

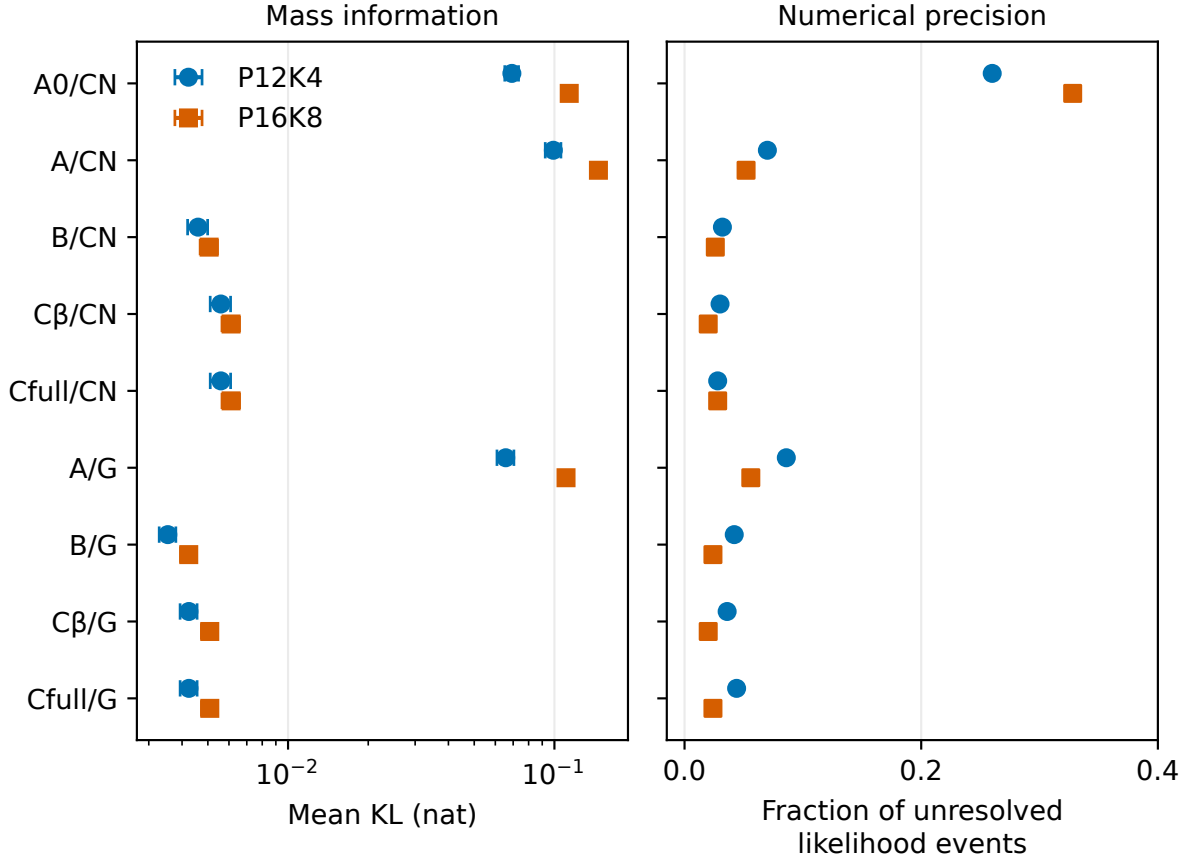


Figure 1: Prior-only information and numerical resolution in the completed campaign. Left: mean posterior-to-prior KL divergence, with one Monte Carlo standard error. Right: fraction of unresolved log-likelihood events among 500 prior draws. P12K4 and P16K8 are shown separately. Under approximate models, KL is descriptive rather than mutual information of the true generator.

Compressed posteriors are weakly informative in this design. Similarity between B and frozen-response controls therefore does not validate the latter across other distances, spectra or signal strengths. Nor can this experiment separate the effect of increasing pulsar count from increasing frequency coverage. Raw complex data are nested, whereas the two sets of angular summaries need not be nested.

The fixed cells provide conditional recovery rather than SBC. At $u = 0$, a central credible interval under a continuous prior structurally excludes the boundary; zero central coverage is not a numerical failure. Count intervals and binomial uncertainty are retained in the machine-readable results, rather than removing difficult realizations.

5 Discussion and limits

The experiments separate two effects: compression reduces information in the correctly specified Gaussian control, while a normal approximation to physical quadratic estimators can fail calibration. This distinction matters for comparisons of dispersive and reference-frequency responses, because a change in a misspecified posterior mixes response effects with the consequences of the approximate likelihood.

An additional conditional comparison uses the first 50 prior realizations of P16K8, selected before computing paired contrasts, with the nuisance parameters fixed. Inverting the order-32 and order-64 CDF envelopes, enlarged vertically by 0.002, gives numerical sensitivity intervals for $Q_{0.95}(u)$. All 200 paired contrasts have numerical deviations below 0.01 in either direction. The numerical intervals for the four sample means, comparing $C_\beta - B$ and $C_{\text{full}} - C_\beta$ in the Gaussian and physical-data families, lie within $[-0.0044, 0.0041]$. This establishes small mean shifts in this conditional sample; the intervals describe numerical sensitivity, not uncertainty in a population mean. The earlier comparison with nuisance-parameter marginalization remains inconclusive at the 0.01 scale.

Low information gain also affects the interpretation of upper limits. For a constant likelihood and a uniform prior on $u \in [0, 1]$, the posterior has $Q_{0.95} = 0.95$ and $D_{\text{KL}} = 0$. The corresponding mass limit $m_g c^2 \leq 0.95h/T$ follows entirely from the assumed support. Comparing the posterior with the prior distinguishes this baseline from a data-driven restriction. Numerical proximity to that baseline alone is insufficient to diagnose a published analysis.

The companion dissertation extends the physical response to linked Fierz–Pauli helicity zero: transverse and longitudinal scalar deformations belong to one amplitude and contribute cross terms to the covariance. It also studies synthetic timing fits. Projection of spin and annual terms generates nonzero pseudocovariance and correlations between Fourier frequencies, even for Gaussian noise. These effects motivate a joint treatment of timing and stochastic parameters when moving from the periodic experiment to irregularly sampled pulse arrival times.

The present conclusions concern simulated arrays with known nuisance parameters. Their extension to observations requires a timing projection, realistic cadence, distance uncertainties and joint noise inference. The results provide a basis for that extension: assess likelihood calibration, quantify information loss, and interpret response-induced shifts relative to both numerical uncertainty and prior sensitivity.

Data and code availability

Code, configurations, numerical results and reproduction instructions are maintained at <https://github.com/flavioluiz/TGdoWayne>. Public sky metadata originate from MeerKAT, DOI 10.57891/j0vh-5g31, under CC BY 4.0. Distances, gravitational signals and noise in the experiments are synthetic.

The population generation and posterior synthesis were repeated using the same pinned numerical environment in a separate directory. The numerical arrays, posterior summaries,

calibration statistics and recovery summaries matched the original calculation. Computational details and the scope of this reproduction are documented with the code.

References

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